

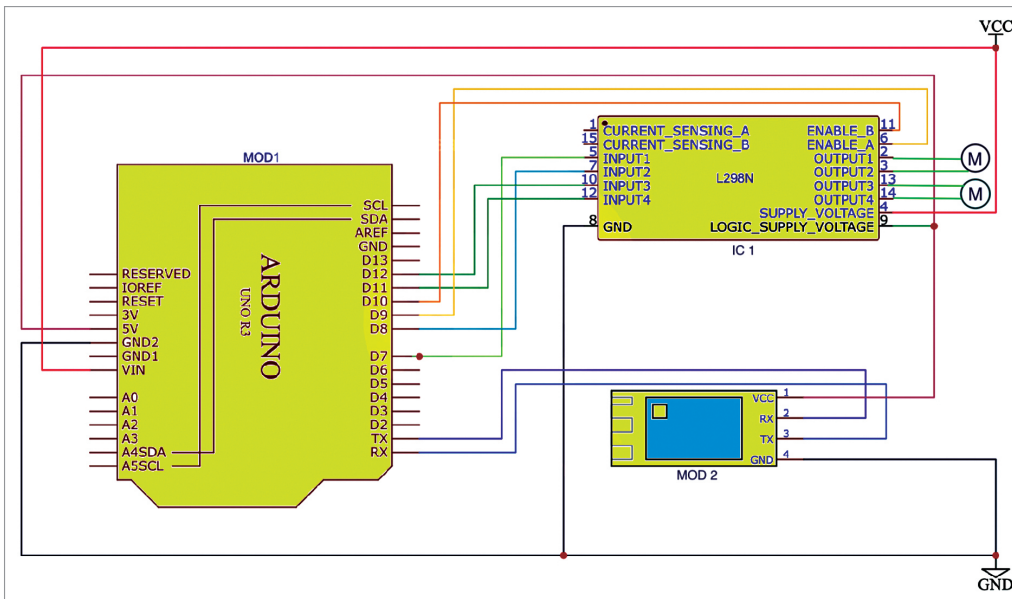


Fig. 2: Circuit diagram

Waterrobot.ino

```

1
2 char junk;
3 String inputString="";
4
5 void setup()                // run once, when the sketch starts
6 {
7   Serial.begin(9600);       // set the baud rate to 9600, same should be of your Serial Monitor
8   pinMode(13, OUTPUT);
9   pinMode(12, OUTPUT);
10  pinMode(11, OUTPUT);
11  pinMode(7, OUTPUT);
12  pinMode(8, OUTPUT);
13  pinMode(9, OUTPUT);
14  pinMode(10, OUTPUT);
15 }
16
17 void loop()
18 {
19   if(Serial.available()){
20     while(Serial.available())
21     {
22       char inChar = (char)Serial.read(); //read the input
23       inputString += inChar;             //make a string of the characters coming on serial
24     }
25     Serial.println(inputString);
26     while (Serial.available() > 0)
27     { junk = Serial.read(); }           // clear the serial buffer
28     if(inputString == "F"){              //in case of 'a' turn the LED on
29       digitalWrite(13, HIGH);
30       digitalWrite(12, LOW);
31       digitalWrite(11, HIGH);
32       analogWrite(9, 255);
33       digitalWrite(7, HIGH);
34       digitalWrite(8, LOW);
35       analogWrite(10, 255);
36     }
37     else if(inputString == "B"){         //incase of 'b' turn the LED off
38       digitalWrite(13, HIGH);
39       digitalWrite(12, HIGH);
40       digitalWrite(11, LOW);
41     }

```

Fig. 3: Code snippet

driver, respectively. The 500-RPM DC motor is attached to output1 and output2 pins 2 and 3, as well as output3 and output4 pins 13 and 14 of the L298 motor driver.

Software

Based on the circuit shown in Fig. 2, the software was developed in the Arduino platform. Arduino Uno has digital pins D2 through D13 and analogue pins A0 through A7. Arduino Uno sends commands to the L298 motor driver, which in turn controls the motors to move forward, backward, left, or right.

Arduino Uno receives commands from an Android mobile app through the HC-05 Bluetooth sensor. The Android app can be downloaded from the Google Play Store.

When the user sends commands like 'f' or 'F' from the mobile app to Arduino Uno, the vehicle moves forward. Sending 'B' or 'b' commands the vehicle to move backward. Commands 'L' or 'l' initiate a left turn, 'R' or 'r' a right turn, and 'S' or 's' makes it stop. This allows for convenient control of the water-serving robot through a mobile device. The robot can be deployed in schools, college